IGCSE 0607 Practice Questions — E7.2 — Combined Questions & Markscheme

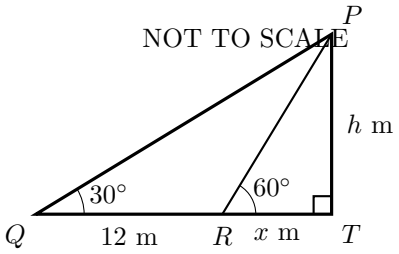
Right-angled triangles

Total Questions: 18 — **Total Marks:** 125 Marks

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- **E7.2** Right-angled triangles
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(17 Qs)*

1 The diagram shows a vertical tower PT of height h metres standing on horizontal ground QRT.
From point Q, the angle of elevation of the top of the tower P is $30°$.
Point R lies on the straight line QT such that $QR = 12$ m and $RT = x$ m. From point R, the angle of elevation of P is $60°$.



(a) Express x in terms of h, and hence show that the height of the tower is $h = 6\sqrt{3}$ m.

[4]

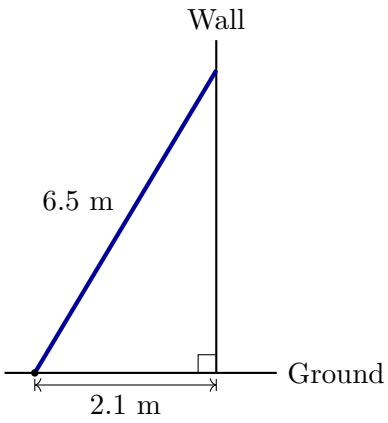
(b) Calculate the exact perimeter of triangle PQR, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

Perimeter = ... m **[3]**

(c) Calculate the exact shortest distance from the base of the tower T to the line PR, giving your answer in the form $k\sqrt{3}$, where k is an integer.

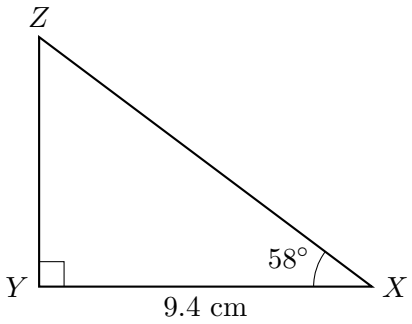
Distance = ... m **[3]**

2 A straight ladder of length 6.5 m leans against a vertical wall.
The base of the ladder is 2.1 m from the bottom of the wall on horizontal ground.



Calculate the angle the ladder makes with the horizontal ground.

..... ° **[3]**



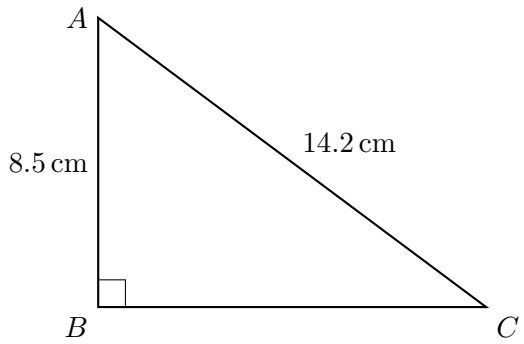
NOT TO SCALE

3 Triangle XYZ is right-angled at Y.
Angle $ZXY = 58°$ and $XY = 9.4$ cm.

Calculate the length of XZ.

..... cm **[3]**

4 The diagram shows a right-angled triangle ABC with the right angle at B.
$AB = 8.5$ cm and $AC = 14.2$ cm.



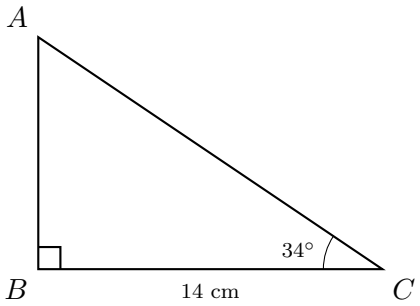
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate angle BAC.

Angle $BAC =$ ..... ° **[2]**

5 The diagram shows a right-angled triangle ABC with $\angle ABC = 90°$, $BC = 14$ cm and $\angle ACB = 34°$.



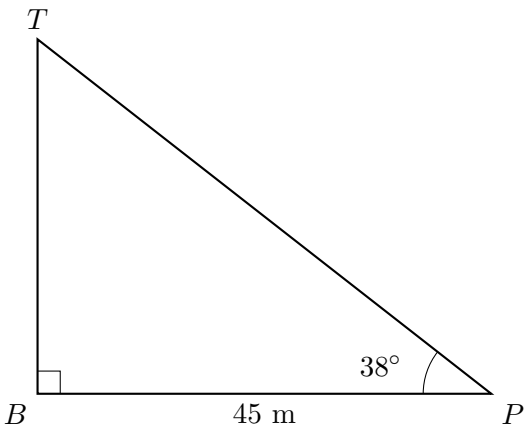
(a) Calculate the length of AB.

$AB =$ ..... cm **[2]**

(b) Point D lies on AC such that BD is the shortest distance from B to AC.
Calculate the length of BD.

$BD =$ ..... cm **[3]**

6 A surveyor stands at a point P, which is 45 m away from the base, B, of a vertical tower TB. The angle of elevation of the top of the tower, T, from P is 38°.



(a) Calculate the height of the tower, TB.

..... m **[2]**

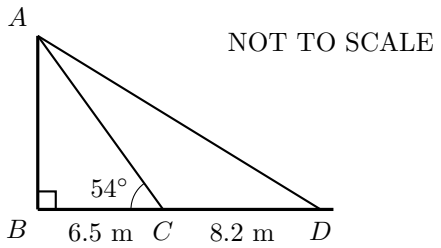
(b) Calculate the straight-line distance PT.

..... m **[2]**

(c) A bird sits at a point X on the tower, where $BX = 20$ m.
Calculate the angle of elevation of the bird from P.

..... ° **[3]**

7 The diagram shows a vertical flagpole AB standing on horizontal ground BCD.
A straight wire AC is attached to the top of the flagpole A and anchored at C, where $BC = 6.5$ m and angle $ACB = 54°$.
A second straight wire AD is anchored at D, where $CD = 8.2$ m.



(a) Calculate the height AB of the flagpole, giving your answer correct to 3 significant figures.

$AB =$ ..... m **[2]**

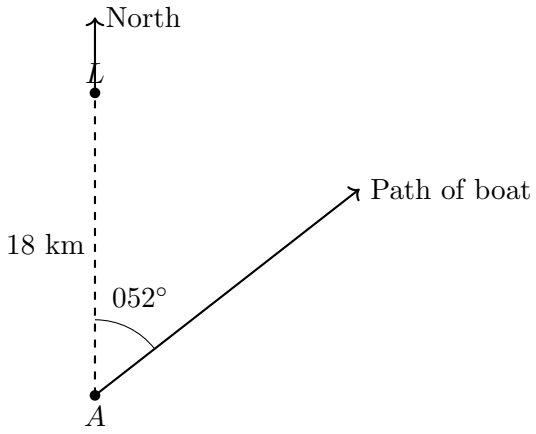
(b) Calculate the angle of depression of point D from the top of the flagpole A, giving your answer correct to 1 decimal place.

..... $°$ **[3]**

(c) Calculate the shortest distance from point B to the wire AD, giving your answer correct to 3 significant figures.

Distance = ... m **[3]**

8 A boat sails from a point A in a straight line on a bearing of 052°.
A lighthouse, L, is located 18 km due North of A.



NOT TO SCALE

(a) Calculate the shortest distance from the lighthouse L to the path of the boat.

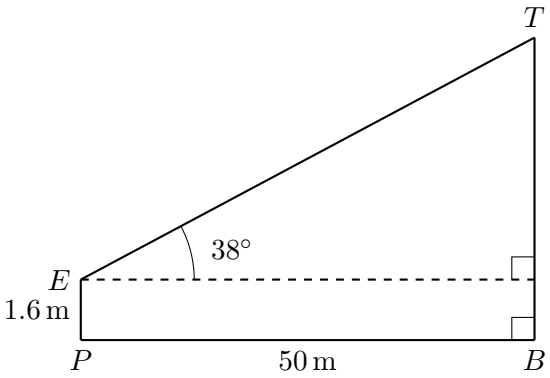
..... km **[2]**

(b) The boat travels at a constant speed of 14 km/h.

Calculate the time it takes for the boat to reach the point on its path closest to the lighthouse.
Give your answer in minutes, correct to the nearest minute.

..... minutes **[3]**

9 A vertical tower TB stands on horizontal ground.
A person stands at point P, 50 m from the base of the tower B.
The person's eyes are at point E, 1.6 m vertically above P.
The angle of elevation of the top of the tower, T, from E is 38°.



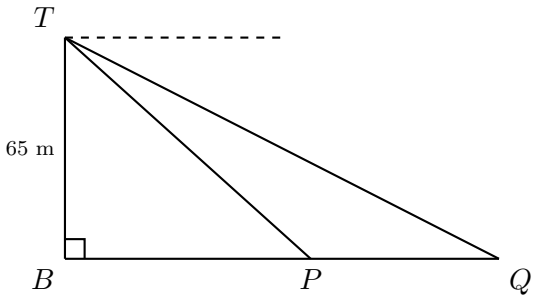
(a) Calculate the total height of the tower, TB.

$TB =$ ..... m **[3]**

(b) A bird on top of the tower at T looks down at a coin on the ground at point X.
The point B lies on the straight line between P and X.
The angle of depression of X from T is 25°.
Calculate the distance BX.

$BX =$ ..... m **[3]**

10 From the top T of a vertical cliff TB of height 65 m, a lighthouse keeper observes two boats, P and Q, at sea.
Points B, P and Q lie in a straight horizontal line with P closer to the base of the cliff B than Q.
The angle of depression of boat P from T is 42° and the angle of depression of boat Q from T is 27°.



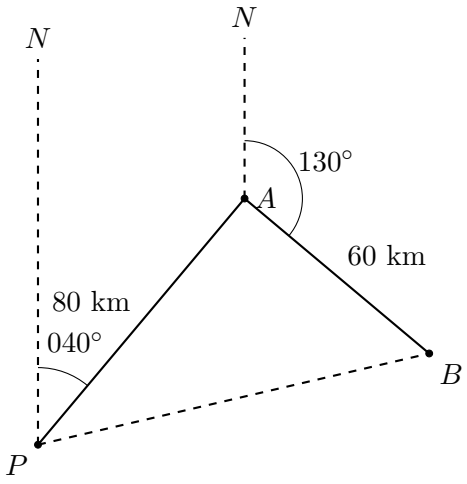
(a) Calculate the distance BP of boat P from the base of the cliff.

$BP =$ ..... m **[2]**

(b) Calculate the distance PQ between the two boats.

$PQ =$ ..... m **[4]**

11 A ship sails from port P on a bearing of 040° for 80 km to point A.
It then changes course and sails on a bearing of 130° for 60 km to point B.



(a) Calculate the direct distance from P to B.

$PB =$ ..... km **[3]**

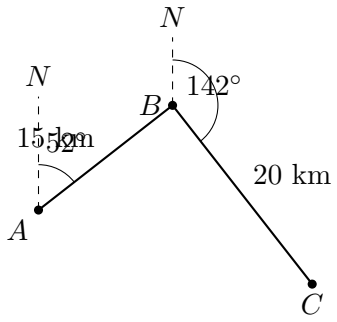
(b) Calculate the bearing of B from P.

Bearing = ... ° **[3]**

(c) Calculate the shortest distance from point A to the path PB.

..... km **[3]**

12 A ship sails from port A to point B on a bearing of 052° for 15 km.
It then sails from point B to point C on a bearing of 142° for 20 km.



(a) Show that angle $ABC = 90°$.

[2]

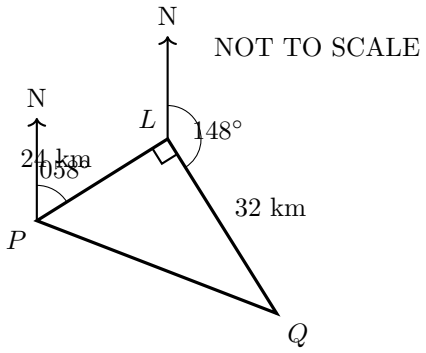
(b) Calculate the distance AC.

..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[4]**

13 A ship sails from port P on a bearing of 058° for 24 km to a lighthouse L.
It then changes course and sails on a bearing of 148° for 32 km to an island Q.



(a) Show that angle $PLQ = 90°$.

[2]

(b) Calculate

(i) the direct distance PQ from the port to the island,

$PQ =$ ..... km **[2]**

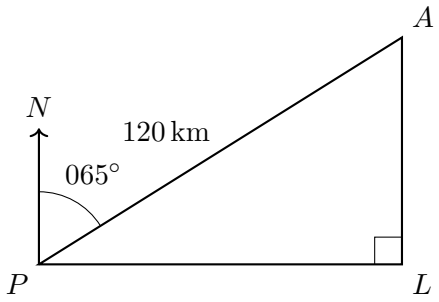
(ii) the bearing of the island Q from the port P, giving your answer correct to 1 decimal place.

..... ° **[3]**

(c) A rescue boat is stationed at lighthouse L.
Calculate the shortest distance from the rescue boat to the ship if the ship sails directly along the straight line PQ.

Distance = ... km **[2]**

14 A ship sails from a port P on a bearing of 065° for 120 km to point A.
A lighthouse is situated at point L, which is due East of P.
Point A is due North of the lighthouse L.



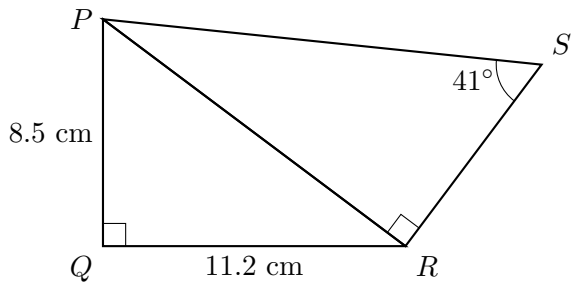
(a) Calculate the distance PL.

$PL =$ ..... km **[3]**

(b) Find the shortest distance from the lighthouse L to the path of the ship PA.

..... km **[3]**

15 The diagram shows a quadrilateral $PQRS$ formed by two right-angled triangles, PQR and PRS.
Angle $PQR = 90°$ and angle $PRS = 90°$.
$PQ = 8.5$ cm, $QR = 11.2$ cm and angle $PSR = 41°$.



NOT TO SCALE

(a) Calculate the length of PR.

..... cm **[2]**

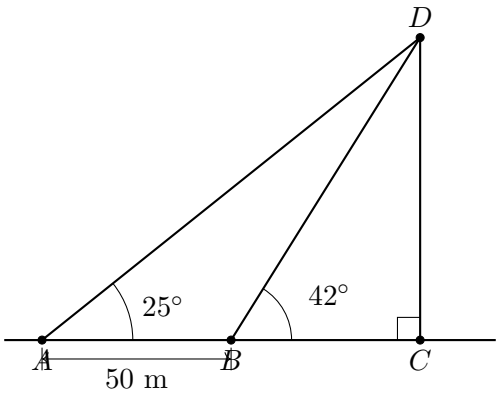
(b) Calculate the length of PS.

..... cm **[2]**

(c) Calculate the perpendicular distance from R to the line PS.

..... cm **[2]**

16 Point A and point B are on horizontal ground, in a straight line with the base C of a vertical building CD.
The distance between A and B is 50 m.
The angle of elevation of the top of the building, D, from A is 25° and from B is 42°.



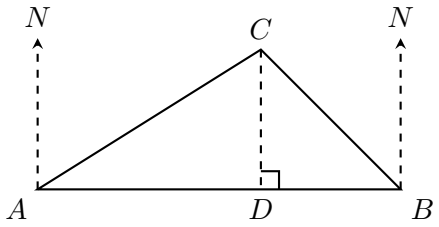
(a) Calculate the height of the building CD.

$CD =$ ..... m **[5]**

(b) A bird flies directly along the straight path from A to D.
Calculate the shortest distance from point B to the flight path AD.

..... m **[3]**

17 Port B is 48 km due east of port A.
A ship sails from port A on a bearing of 058° until it reaches a point C which is directly north-west of port B (on a bearing of 315° from B).
Point D lies on the line AB such that CD is perpendicular to AB.



(a) Write down the size of angle CAB and the size of angle CBA.

$\angle CAB =$ ..... , $\angle CBA =$ ..... **[2]**

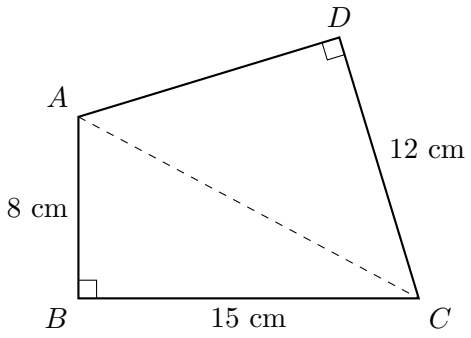
(b) Calculate the shortest distance, CD, of the ship from the straight line path AB.

$CD =$ ..... km **[4]**

(c) Calculate the distance AC sailed by the ship from port A to point C.

$AC =$ ..... km **[2]**

18 The diagram shows a quadrilateral $ABCD$.
Angle $ABC = 90°$ and angle $ADC = 90°$.
$AB = 8$ cm, $BC = 15$ cm, and $CD = 12$ cm.



(a) Calculate the length of AC.

..... cm **[2]**

(b) Calculate the length of AD.

..... cm **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

(d) Calculate the size of angle DAB.

..... ° [4]

1 **(a)** $\tan 60° = \dfrac{h}{x}$ giving $x = \dfrac{h}{\sqrt{3}}$ **[M1]**

$\tan 30° = \dfrac{h}{x+12}$ giving $\dfrac{1}{\sqrt{3}} = \dfrac{h}{\frac{h}{\sqrt{3}}+12}$ **[M1]**

$\dfrac{h}{\sqrt{3}} + 12 = h\sqrt{3}$ multiplying by $\sqrt{3}$ gives $h + 12\sqrt{3} = 3h$ **[M1]**

$2h = 12\sqrt{3}$ giving $h = 6\sqrt{3}$ m **[A1]**

(b) $x = 6$ m and $PR = \dfrac{6\sqrt{3}}{\sin 60°} = 12$ m **[B1]**

$PQ = \dfrac{18}{\cos 30°} = \dfrac{36}{\sqrt{3}} = 12\sqrt{3}$ m **[M1]**

Perimeter $= 12 + 12 + 12\sqrt{3} = 24 + 12\sqrt{3}$ **[A1]**

(c) $\dfrac{1}{2} \times PR \times d = \dfrac{1}{2} \times RT \times PT$ (or $d = 6\sin 60°$) **[M1]**

$d = \dfrac{6 \times 6\sqrt{3}}{12}$ **[M1]**

$3\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

2 Identifies cosine ratio: $\cos(\theta) = \dfrac{2.1}{6.5}$ **[M1]**

$\theta = \cos^{-1}\left(\dfrac{2.1}{6.5}\right)$ **[M1]**

71.2 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 $\cos(58°) = \dfrac{9.4}{XZ}$ **[M1]**

$XZ = \dfrac{9.4}{\cos(58°)}$ **[M1]**

17.7 or 17.73... **[A1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** $BC^2 + 8.5^2 = 14.2^2$ **[M1]**

$BC = \sqrt{14.2^2 - 8.5^2}$ OR $BC = \sqrt{129.39}$ **[M1]**

11.4 (accept 11.37...) **[A1]**

(b) $\cos(\angle BAC) = \dfrac{8.5}{14.2}$ **[M1]**

53.2 (accept 53.23...) **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Uses tangent ratio: $\tan(34°) = \dfrac{AB}{14}$ or $AB = 14\tan(34°)$ **[M1]**

9.44 (accept 9.443) **[A1]**

(b) Recognises shortest distance is perpendicular to AC, so $\angle BDC = 90°$ **[B1]**

Uses sine ratio in right-angled triangle BDC: $\sin(34°) = \dfrac{BD}{14}$ **[M1]**

7.83 (accept 7.828 to 7.83) **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $\tan(38°) = \dfrac{TB}{45}$ **[M1]**

35.2 (or 35.15 to 35.16) **[A1]**

(b) $\cos(38°) = \dfrac{45}{PT}$ or $PT^2 = 45^2 + 35.15^2$ **[M1]**

57.1 (or 57.10 to 57.11) **[A1]**

(c) $\tan(XPB) = \dfrac{20}{45}$ **[M1]**

Angle $= \tan^{-1}\left(\dfrac{20}{45}\right)$ **[M1]**

24.0 (or $23.96\ldots$) **[A1]**

[Back to Question] *[Total Marks: 7]*

7 **(a)** $\tan 54° = \dfrac{AB}{6.5}$ giving $AB = 6.5\tan 54°$ $(= 8.946\ldots)$ **[M1]**

8.95 **[A1]**

(b) $BD = 6.5 + 8.2 = 14.7$ m **[B1]**

$\tan\angle ADB = \dfrac{8.946\ldots}{14.7}$ $(= 0.6086\ldots)$ **[M1]**

31.3 **[A1]**

(c) Shortest distance $d = 14.7\sin 31.32\ldots°$ (or $AD = \sqrt{8.946\ldots^2 + 14.7^2} = 17.208\ldots$ m) **[M1]**

$d = 7.642\ldots$ (or $\dfrac{8.946\cdots \times 14.7}{17.208\ldots}$) **[M1]**

7.64 **[A1]**

[Back to Question] *[Total Marks: 8]*

8 **(a)** Uses $\sin(52°) = \dfrac{\text{distance}}{18}$ or $18 \times \sin(52°)$ **[M1]**

14.18... or 14.2 **[A1]**

(b) Distance to closest point $= 18 \times \cos(52°) = 11.082...$ km **[M1]**

Time $= \dfrac{11.082...}{14}$ $(= 0.7915...$ hours) **[M1]**

$0.7915... \times 60 = 47.49... = 47$ **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** $\tan(38°) = \dfrac{h}{50}$ **[M1]**

$h = 50\tan(38°)$ OR $39.06\ldots$ **[A1]**

Total height $= 39.06\cdots + 1.6 = 40.7$ (accept $40.66\ldots$) **[A1]**

(b) Identifies the angle of elevation of T from X as $25°$ OR $\angle BTX = 65°$ **[B1]**

$\tan(25°) = \dfrac{40.66}{BX}$ OR $\tan(65°) = \dfrac{BX}{40.66}$ **[M1]**

$BX = 87.2$ (accept $87.19\ldots$) **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** Uses $\angle TPB = 42°$ (or $\angle BTP = 48°$): $\tan(42°) = \dfrac{65}{BP}$ or $BP = 65\tan(48°)$ **[M1]**

72.2 (accept 72.18 to 72.2) **[A1]**

(b) Uses $\angle TQB = 27°$ (or $\angle BTQ = 63°$): $\tan(27°) = \dfrac{65}{BQ}$ **[M1]**

Calculates $BQ = \dfrac{65}{\tan(27°)} = 127.57...$ **[A1]**

Subtracts distances: $PQ = 127.57... - 72.189...$ **[M1]**

55.4 (accept 55.38 to 55.4) **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Calculates $\angle PAB = 360° - (180° - 40°) - 130° = 90°$ (or similar valid reasoning to show right angle) **[M1]**

Uses Pythagoras' theorem: $PB^2 = 80^2 + 60^2$ **[M1]**

100 **[A1]**

(b) Uses trigonometry to find $\angle APB$: $\tan(APB) = \dfrac{60}{80}$ **[M1]**

$\angle APB = \tan^{-1}(0.75) = 36.869...°$ **[A1]**

Bearing $= 040° + 36.869...° = 076.9$ (accept 077) **[A1]**

(c) Recognises the shortest distance d is the perpendicular height of $\triangle PAB$ from A to PB. **[M1]**

Uses area equation: $\dfrac{1}{2} \times 100 \times d = \dfrac{1}{2} \times 80 \times 60$ OR uses trigonometry $80\sin(53.13°)$ or $60\sin(36.87°)$ **[M1]**

48 **[A1]**

[Back to Question] *[Total Marks: 9]*

12 **(a)** Interior angle between parallel North lines $= 180° - 52° = 128°$ **[M1]**

Angles around point $B = 360° - 128° - 142° = 90°$ with no errors seen **[A1]**

(b) $AC = \sqrt{15^2 + 20^2}$ **[M1]**

$AC = \sqrt{625}$ **[M1]**

25 **[A1]**

(c) $\tan(BAC) = \dfrac{20}{15}$ **[M1]**

Angle $BAC = 53.13°$ **[A1]**

Bearing $= 052° + 53.13°$ **[M1]**

105.1 (or $105.13\ldots$) **[A1]**

[Back to Question] *[Total Marks: 9]*

13 **(a)** Bearing of P from $L = 058° + 180° = 238°$ **[B1]**

Angle $PLQ = 238° - 148° = 90°$ **[A1]**

(b) **(i)** $PQ^2 = 24^2 + 32^2 = 576 + 1024 = 1600$ **[M1]**

40 **[A1]**

(ii) $\tan \angle LPQ = \dfrac{32}{24}$ giving angle $LPQ = 53.130\ldots°$ **[M1]**

Bearing $= 058° + 53.130\ldots°$ **[M1]**

111.1 **[A1]**

(c) $\dfrac{1}{2} \times 40 \times d = \dfrac{1}{2} \times 24 \times 32$ (or $d = 24 \sin 53.130\ldots°$) **[M1]**

19.2 **[A1]**

[Back to Question] *[Total Marks: 9]*

14 **(a)** $\angle APL = 90° - 65° = 25°$ **[B1]**

$\cos(25°) = \dfrac{PL}{120}$ OR $\sin(65°) = \dfrac{PL}{120}$ **[M1]**

109 or 108.8 (accept $108.75\ldots$) **[A1]**

(b) Identifies the shortest distance as the perpendicular distance from L to PA, creating a new right-angled triangle with hypotenuse PL **[B1]**

$\sin(25°) = \dfrac{d}{108.76}$ OR Area method: $0.5 \times 120 \times d = 0.5 \times 108.76 \times (120 \sin 25°)$ **[M1]**

46.0 (accept $45.96\ldots$) **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** $PR = \sqrt{8.5^2 + 11.2^2}$ or $\sqrt{197.69}$ **[M1]**

14.06... or 14.1 **[A1]**

(b) $\sin(41°) = \dfrac{14.06...}{PS}$ or $PS = \dfrac{14.06...}{\sin(41°)}$ **[M1]**

21.43... or 21.4 **[A1]**

(c) Angle $RPS = 180° - 90° - 41° = 49°$ AND distance $= 14.06... \times \sin(49°)$
OR area method: $0.5 \times 21.43... \times h = 0.5 \times 14.06... \times (14.06.../\tan(41°))$ **[M1]**

10.60... or 10.6 **[A1]**

[Back to Question] *[Total Marks: 6]*

16 (a) Sets up equation for $\triangle BCD$: $\tan(42°) = \dfrac{CD}{BC}$

BC $= \dfrac{CD}{\tan(42°)}$ [M1]

Sets up equation for $\triangle ACD$: $\tan(25°) = \dfrac{CD}{BC + 50}$

BC + 50 $= \dfrac{CD}{\tan(25°)}$ [M1]

Equates or substitutes to eliminate BC: $50 = \dfrac{CD}{\tan(25°)} - \dfrac{CD}{\tan(42°)}$ **[M1]**

Factorises to solve for CD: $CD\left(\dfrac{1}{\tan(25°)} - \dfrac{1}{\tan(42°)}\right) = 50$ **[M1]**

48.4 (accept 48.36...) **[A1]**

(b) Recognises the shortest distance d is the perpendicular from B to AD, forming a right-angled triangle with hypotenuse $AB = 50$. **[M1]**

Sets up trigonometric ratio: $\sin(25°) = \dfrac{d}{50}$ **[M1]**

21.1 (accept 21.13...) **[A1]**

[Back to Question] *[Total Marks: 8]*

17 (a) $\angle CAB = 90° - 58° = 32°$ **[B1]**

$\angle CBA = 315° - 270° = 45°$ **[B1]**

(b) Expresses AD and BD in terms of CD: $AD = \dfrac{CD}{\tan(32°)}$ and $BD = \dfrac{CD}{\tan(45°)} = CD$ **[M1]**

Sets up sum equation: $\dfrac{CD}{\tan(32°)} + CD = 48$ **[M1]**

Factors out CD: $CD\left(\dfrac{1}{\tan(32°)} + 1\right) = 48$ giving $CD(2.6003...) = 48$ **[M1]**

18.5 (accept 18.45 to 18.46) **[A1]**

(c) Uses sine ratio in triangle ACD: $\sin(32°) = \dfrac{18.459...}{AC}$ **[M1]**

34.8 (accept 34.83 to 34.9) **[A1]**

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18 **(a)** $AC = \sqrt{8^2 + 15^2}$ **[M1]**

17 **[A1]**

(b) $AD^2 + 12^2 = 17^2$ (or their AC^2) **[M1]**

$AD = \sqrt{289 - 144} = \sqrt{145}$ **[M1]**

12.0 (or 12.04...) **[A1]**

(c) Area of $\triangle ADC = \frac{1}{2} \times 12 \times \sqrt{145} = 72.25\ldots$ **[M1]**

$\frac{1}{2} \times 17 \times h = 72.25\ldots$ **[M1]**

8.50 (or 8.499...) **[A1]**

(d) $\tan(BAC) = \frac{15}{8} \implies$ Angle $BAC = 61.93°$ **[M1]**

$\tan(CAD) = \frac{12}{\sqrt{145}} \implies$ Angle $CAD = 44.90°$ **[M1]**

Angle $DAB = 61.93° + 44.90°$ **[M1]**

106.8 (or 106.82 to 106.83) **[A1]**

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